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系统性硬化症的表观遗传学研究进展

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[摘要] 系统性硬化症(systemic sclerosis，SSc)是一类病因不明的自身免疫性疾病，以血管病变、炎症和皮肤及各器官广泛纤维化为主要特征。纤维化是SSc的标志，也是导致其高病死率的主要原因。近年来，SSc的表观遗传学(DNA甲基化、组蛋白修饰和非编码RNA)得到了深入的研究，其中以DNA甲基化和微小RNA(microRNA，miRNA)的研究最为广泛。DNA甲基化异常可影响SSc的血管内皮细胞、$CD4^+$T细胞和成纤维细胞的功能。血清中的miRNAs与自身抗体、SSc疾病活动度和并发症密切相关，成纤维细胞中的miRNAs可直接影响成纤维细胞的活化。

[关键词] 系统性硬化症；表观遗传学；DNA甲基化；组蛋白修饰；微小RNAs；长链非编码RNA

Research progress in epigenetic studies on systemic sclerosis

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ABSTRACT Systemic sclerosis (SSc) is an autoimmune disease with unknown etiology, characterized by vasculopathy, inflammation, and extensive fibrosis in the skin and organs. Fibrosis is the hallmark of SSc and contributes to its high mortality. In recent years, with the in-depth study of the epigenetics of SSc (DNA methylation, histone modification, and non-coding RNA), the DNA methylation and miRNA has been the most widely studied. Abnormal DNA methylation can influence the function of vascular endothelial cells, $CD4^+$ T cells, and fibroblasts in SSc. MiRNAs in serum is closely related to autoantibodies, SSc disease activity and complications, and miRNAs in fibroblasts can directly affect the activation of fibroblasts.

KEY WORDS systemic sclerosis; epigenetic; DNA methylation; histone modification; microRNAs; long non-coding RNA

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系统性硬化症(systemic sclerosis，SSc)是一种病因不明的多系统结缔组织病，以自身免疫、炎症、血管结构和功能异常以及皮肤和各脏器进行性纤维化为特征[1]。其发病率仅次于类风湿关节炎和系统性红斑狼疮，是致死率最高的结缔组织病[2]。全身多器官进行性纤维化是SSc高病死率的主要原因。其发病机制尚不清楚，目前认为在遗传和环境的作用下，自身免疫和炎症反应、血管病变、纤维化三者相互影响而形成恶性循环，最终导致各器官进行性纤维化[3]。根据皮肤受累程度SSc分为弥漫性系统性硬化症(diffuse cutaneous scleroderma，dSSc)和局限性系统性硬化症(limited cutaneous scleroderma，lSSc)。dSSc有广泛的皮肤受损，由四肢近端延伸到肘部和膝部，常累及胸壁和腹壁，病情进展快，常在首发症状出现后5年内出现脏器衰竭。lSSc皮肤受损主要局限于肢体末端或面部，或仅累及手指(指端硬化)。病情进展缓慢，预后较好[4]。

目前，通过国际多中心全基因组关联研究(genome-wide association study，GWAS)已筛选出大量SSc易感基因，如人类白细胞抗原-DPB1(human leukocyte antigen-DPB1，HLA-DPB1)，HLA-DPB2，HLA-DQB1，干扰素调节因子5[interferon regulatory factor (IRF) 5，IRF5]、信号转导与转录激活因子4(signal transducers and activators of tranion 4，STAT4)和自噬相关基因5(autophagy related gene5，ATG5)等，这些基因广泛参与炎症、T细胞分化和细胞自噬等[5]。然而，双生子对照研究[6]发现遗传易感性仅占SSc发病的4.2%，提示环境-表观遗传学可能在SSc发病中起着更重要的作用。表观遗传学是指在基因核苷酸序列不发生改变的情况下，基因表达的可遗传的变化，主要包括DNA甲基化、组蛋白修饰、微小RNA(microRNA，miRNA)和长链非编码RNA(long non-coding RNA，lncRNA)等[7]。近年来，作者所在课题组中南大学湘雅医院风湿免疫科利用高通量芯片技术对SSc进行DNA甲基化和miRNA表达谱分析，并深入探讨了相关机制。本文综合国内外重要研究进展及课题组前期的研究发现，为阐明SSc的发病机制、寻找特异的生物标志物及潜在治疗靶点提供参考。

1 DNA甲基化

DNA甲基化是指在DNA甲基化转移酶(DNA methyltransferase，DNMTs)的作用下将甲基选择性地添加到胞嘧啶上形成5-甲基胞嘧啶的过程。DNA高甲基化通常抑制基因表达，去甲基化则诱导基因的重新活化和表达[8]。

Selmi等[9]分析8对女性同卵双生子(其中7对双生子中每对有1人患SSc，另1对1人为dSSc，1人为lSSc)外周血单个核细胞(peripheral blood mononuclear cell，PBMC)X染色体甲基化水平时发现：18个高甲基化位点和25个低甲基化位点与SSc发病相关，这些甲基化异常的基因包括转录因子[无芒相关同源框(aristaless-related homeobox，ARX)，热休克转录因子X1(heat shock transcription factor, X-linked，HSFX1)，含BED型锌指蛋白1(zinc finger BED-type containing 1，ZBED1)和锌指蛋白41(zinc finger protein 41，ZNF41)]、表面抗原[白介素1受体辅助蛋白样蛋白2(interleukin 1 receptor accessory protein-like 2，IL1RAPL2)和孕酮受体膜组分1基因(progesterone receptor membrane component 1，PGRMC1)]、细胞增殖因子[磷酸甘油酸激酶1(phosphoglycerate kinase 1，PGK1)，精胺合成酶(spermine synthase，SMS)，U3小核仁RNA相关蛋白14A (U3 small nucleolar RNA-associated protein 14 homolog A，UTP14A)和重组人信号序列受体4(signal sequence receptor 4，SSR4)]与凋亡相关基因[如肌微管素1(myotubularin 1，MTM1)]、炎症相关基因[如丝氨酸/苏氨酸激酶A-Raf(serine/threonine-protein kinase A-Raf，ARAF)]和氧化应激相关基因[如胞外吡啶核苷酸氧化酶二硫键巯基交换酶2(ecto-NOX disulfide-thiol exchanger 2，ENOX2)]等。此外，SSc患者的PBMC中还存在IRF7低甲基化[10]。对SSc患者的$CD4^+$和$CD8^+$T细胞进行全基因组甲基化测序发现：I型干扰素信号通路基因明显低甲基化[11]。在SSc $CD4^+$T细胞中，DNMT1，重组人甲基化CpG结合蛋白3(methyl-CpG-binding domain protein 3，MBD3)和MBD4基因表达降低[12]，肿瘤坏死因子配体超家族成员7(tumor necrosis factor ligand superfamily member 7，TNFSF7)[13]、白细胞分化抗原40(cluster of differentiation 40，CD40L)[14]和整合素αL亚甲基 (integrin subunit alpha L，TGAL)[15]基因低甲基化，而叉头状螺旋转录因子3(forkhead box P3，FOXP3)[16]高甲基化。作者课题组在前期研究(结果未发表)中利用高通量芯片技术对SSc PBMC进行DNA甲基化、mRNA和细胞因子表达谱整合研究，发现SSc中4 848个低甲基化位点和216个高甲基化位点，1 023个表达上调的基因和1 284个表达下调的基因，384个(57.83%)表达上调的基因甲基化水平降低，14个(2.41%)表达下调的基因甲基化水平增高。一些与自身免疫密切相关的基因如同种异体移植炎症因子1(allograft inflammatory factor 1，AIF1)、AT丰富结合域3A(AT-rich interactive domain-containing protein 3A，ARID3A)、干扰素诱导蛋白44样蛋白(interferon-inducible-protein44，IFI44L)、多聚(ADP核糖)聚合酶9[poly (ADP-ribose) polymerase 9，PARP9]和干扰素诱导基因S-腺苷甲硫氨酸基区域蛋白2(radical SAM

domain-containing 2，RSAD2)等均存在低甲基化和基因高表达。Wnt信号通路和趋化因子信号通路中多个基因存在低甲基化。细胞因子γ干扰素(interferonγ，IFN-γ)、IL-1RA、干扰素γ诱导蛋白10(IFN-γ-inducible protein 10， IP-10)、单核细胞趋化蛋白-1(monocyte chemotactic protein-1，MCP-1)、正常T细胞表达和分泌因子(regulated on activation, normal T cell expressed and secreted，RANTES)和血管内皮生长因子(vascular endothlial growth factor，VEGF)在SSc中表达增高，IL-2和IL-5表达降低。IL-2的甲基化位点chr4:123597758和chr4:123374885呈高甲基化，ILRA-1的甲基化位点chr2:113885277，chr2:113874068和chr2:113885116，VEGF的甲基化位点chr6:43737749均呈低甲基化。

SSc普遍存在微血管病变，在组织活检标本中可发现内皮细胞凋亡现象。微血管内皮细胞凋亡与骨形态发生蛋白II型受体(bone morphogenetic protein Type II receptor，BMPRII)基因密切相关，BMPRII基因上共发现20个高甲基化胞嘧啶-磷酸-鸟嘌呤二核苷酸序列(CpG)位点，有5个位于转录因子结合区域，2个参与BMPRII基因转录[17]。此外，Friend型白血病整合1转录因子(friend leukemia integration 1 transcription factor，FLI1)基因高甲基化使其在SSc皮肤和成纤维细胞中表达降低，FLI1低表达导致血清中趋化因子(C-X-C motif chemokine 5，CXCL5)、血小板源性生长因子(platelet derived growth factor，PDGF-β)和血管内皮细胞(human vascular endothelial cell，VE)钙黏素降低[18]，影响SSc血管内皮细胞的完整性。

在SSc皮肤原代成纤维细胞中，全基因组甲基化芯片发现dSSc中2 710个CpG位点以及lSSc中1 021个CpG位点甲基化水平改变。203个甲基化位点在dSSc和lSSc中共同发生变化，包含118个低甲基化和6个高甲基化基因，这些低甲基化基因与细胞外基质-受体相互作用和黏合斑相关。纤维化相关基因整合素α9(integrin alpha-9，ITGA9)、解整合素金属蛋白酶12(disintegrin and metalloproteinase domain-containing protein 12，ADAM12)、胶原蛋白类型XXIIIα1[collagen α1 (XXIII) chain，COL23A1]、胶原蛋白类型IV α2 [collagen alpha-2(IV) chain，COL4A2]和肌球蛋白IE(myosin-Ie，MYO1E)，以及转录因子Runt 相关转录因子1(runt-related transcription factor 1，RUNX1)、RUNX2和RUNX3均为低甲基化[19]。此外，1型多聚ADP核糖合成酶[poly (ADP-ribose) polymerase 1，PARP1]，FLI1，Krueppel样因子5(Krueppel-like factor 5，KLF5)，重组人Dickkopf相关蛋白1(Dickkopf-related protein 1，DKK1)，人分泌型卷曲相关蛋白1(secreted frizzled-related protein 1，SFRP1)和细胞因子信号转导抑制因子3(suppressor of cytokine signaling 3，SOCS3)基因的高甲基化参与SSc成纤维细胞的活化。PARP1能增强TGF-β信号通路[20]，FLI1通过特异蛋白1(specific protein 1，SP-1)依赖的信号通路负性调节细胞外基质[21-23]，KLF5可提高成纤维细胞的敏感性[24]，DKK1和SFRP1可活化Wnt信号通路[25]，SOCS3基因高甲基化通过活化蛋白酪氨酸激酶2/信号转导子与激活子3(Janus kinase signal transducers 2 and activator of transcription 3，JAK2/STAT3)信号通路参与成纤维细胞活化[26]。

2 组蛋白修饰

组蛋白是真核生物细胞中一类高度保守的蛋白质，以二聚体或四聚体形式与染色质形成核小体。目前有5种不同的组蛋白，分为核心组蛋白(H2A，H2B，H3和H4)和连接组蛋白(H1和H5)。组蛋白N端能被不同的化学修饰，如乙酰化、甲基化和磷酸化，这些修饰可影响染色质与转录因子在启动子区和增强子区的结合。组蛋白乙酰化受到两种作用相反的组蛋白修饰酶的严格调控，分别是组蛋白乙酰转移酶(histone acetyltransferase，HAT)和脱乙酰酶(histone deacetylases，HDAC)。而甲基化受组蛋白甲基转移酶(histone methyltransferases，HMTs)和去甲基化酶调控(histone demethylated，HDMs)[27-28]。在SSc B细胞中，含有组蛋白去甲基化酶2A的Jmjc结合域(Jmjc domain-containing histone demethylase 2A，JHDM2A)表达增高，花斑抑制因子同源物(suppressor of variegation 3-9 homologue 2，SUV39H2)、乙酰转移酶HDAC2和HDAC7表达降低，全基因组组蛋白H4高乙酰化和H3K9低甲基化。HDAC2 蛋白与SSc皮肤厚度、组蛋白H4乙酰化与SSc疾病活动性均呈正相关[29]。在SSc $CD4^+$细胞中，组蛋白去甲基化酶含十字形结构域蛋白3(jumonji domain containing protein 3，JMJD3)表达增高，组蛋白H3第27位赖氨酸三甲基化(histone H3 lysine 27 trimethylation，H3K27me3)表达降低[30]。在SSc成纤维细胞中，H3K27me3抑制剂(3-deazaneplanocin A，3-DZNep)可上调促纤维化转录因子Fos相关抗原(Fos-related antigen-2，Fra-2)和胶原的产生，最终加速TGF-β诱导的纤维化[31]。组蛋白去乙酰酶HDAC7抑制剂曲古抑菌素A (trichostatin A，TSA)可减少I型胶原和纤连蛋白的表达。HDAC7 siRNA可降低I型和III型胶原的表达。在博来霉素诱导的纤维化小鼠中，TSA可减轻皮肤细胞外基质的沉积[23, 32-33]。胶原抑制基因FLI1启动子区组蛋白H3和H4低乙酰化能增加胶原的合成[23]。

3 非编码RNA

3.1 MiRNAs

MiRNAs是一类由真核生物内源基因编码的长约19~25 nt的小分子单链RNA，在进化上具有高度的保守性，在转录后水平调控基因表达。成熟miRNA通过碱基互补配对的方式与靶基因3'端非翻译区(untranslated region，UTR)完全或不完全互补结合，通过诱导mRNA的降解或抑制其翻译而调控靶基因的表达[34]。目前已发现人类基因组中至少能编码800种独立的miRNAs，调控人类近1/3的基因。MiRNA在多种病理、生理过程中发挥着重要作用，如干细胞发育、细胞分化与器官生成、细胞增殖与凋亡、免疫调节等[35]。现已发现了40余种miRNAs通过调控细胞外基质蛋白、基质金属蛋白酶、结缔组织生长因子等纤维化相关基因的表达参与纤维化的发生和发展[36]。

在SSc患者血清中，miRNA-223，-181b，-342-3p和-184可用于区分dSSc和lSSc；miRNA-409，-184，-92a，-29a和-101与抗体及疾病的预后相关。在lSSc中miR-223明显增高，伴miR-409增高和miR-101降低时与抗着丝点抗体(anti-centromere antibody，ACA)相关，而伴miR-29a增高和miR-92a降低时则与抗核糖核蛋白抗体(ribonucleoprotein，RNP)相关。在dSSc中miR-184和miR-181b明显增高，伴miR-92a增高和miR-101降低时与抗甲状腺过氧化物酶抗体(anti-thyroid peroxidase antibody，ATA)相关，而伴miR-342-3p降低则与抗网硬蛋白抗体(anti-gastric parietal cell antibody，ARA)相关[37]。Steen等[38]根据血清中的miRNAs建立了一个含21个miRNAs的模型可判断正常对照、SSc和系统性红斑狼疮(准确率80%)，血清miR-16，-223和-638能区分SSc和正常对照者；miR-142-3p，-150，-223和-638能区分SSc和系统性红斑狼疮，血清miR-638与Scl-70抗体相关。血清中miR-196a水平的降低与dSSc:lSSc的比值、改良的Rodnan皮肤厚度评分(modified Rodnan skin thickness score，mRSS)以及凹陷性瘢痕的发生率升高有关[39]。血清miR-206和miR-21在SSc中明显增高[40]。在SSc早期血清中miR-29a水平显著降低，且与右心室收缩压呈负相关，提示miR-29a可能在肺动脉高压的发病机制中起重要作用[41]。

在SSc皮肤组织中，dSSc和lSSc均具有不同的miRNA表达谱，大多数差异表达的miRNAs与SSc纤维化密切相关[42]。在SSc成纤维细胞和TGF-β诱导的皮肤成纤维细胞中，56个miRNAs共同表达变化(49个表达上调和7个表达下调)[39]。miR-29a能直接抑制纤维化相关基因(胶原、纤连蛋白、弹性蛋白等)的表达，减少细胞外基质的合成而发挥抗纤维化作用。TGF-β和PDGF-β可下调miR-29a的表达而促进心、肝、肺和皮肤纤维化[43-46]。MiR-155通过上调经典的Wnt和Akt信号通路促进成纤维细胞的活化[47-48]。MiR-150[49]，miR-206[42]，miR-129-5P[50]，miR-7[51]，let7a[52]，miR-196a[39, 53]，miR-135b[54]和miR-92a[55]均在SSc纤维化中发挥重要的调控作用。

作者课题组在前期研究[36, 56]中发现：与正常对照组相比较，dSSc皮肤组织中发生变化的miRNAs共42个，其中25个表达上调，17个表达下调。lSSc皮肤组织中发生变化的miRNAs共60个，其中42个表达上调，18个表达下调。在dSSc与lSSc中表达均发生变化的miRNAs共21个，其中13个表达上调，8个表达下调。深入研究其机制发现：miR-21通过调控TGF-β信号通路负性调节蛋白Smad7[57]，miR-130b通过调控过氧化物增殖物受体γ(peroxisome proliferators-activated receptor γ，PPARγ)[58]，miR-202-3P 通过调控基质金属蛋白酶1(matrix metalloproteinase，MMP-1)共同参与SSc的疾病进展[59]。对系统性红斑狼疮、SSc和炎性肌病血清中的miRNAs表达进行分析，发现miR-21在SSc血清中呈特异性表达增高[60]。因此，促纤维化的miRNAs表达增高或抗纤维化的miRNAs表达降低可能在SSc的发病过程中起着重要作用，这些miRNAs将为SSc的治疗提供新的策略。

3.2 LncRNA

LncRNA是一类长度超过200个核苷酸的非编码RNA，通过与DNA，mRNA和蛋白质等相互作用从多水平调控基因表达，如组蛋白修饰、DNA甲基化的调控、转录活性调控和mRNA 的剪辑等，在遗传印记、剂量补偿、细胞周期调控、多能分化、端粒长度调控等方面发挥重要作用。lncRNAs通过调节免疫和炎症在风湿病中也得到了深入研究[61-62]。

Xist基因的反义序列在SSc血清及皮肤成纤维细胞中表达增加，其可能与TGF-β信号通路有关。TSIX siRNA能减少SSc成纤维细胞中I型胶原的表达，过表达TSIX则增加I型胶原的表达。TSIX的表达与康复状态量表(mRSS)呈正相关。因此，TSIX可能成为SSc的生物标志物和治疗靶点[63]。lncRNA MIR503宿主基因在SSc皮肤组织中表达增高。在成纤维细胞、肺血管平滑肌细胞中，TGF-β可诱导lncRNA MIR503HG的生成。使用lncRNA MIR503HG抑制剂SD208和SB31532可显著降低I型胶原、纤连蛋白和α平滑肌动蛋白的表达。lncRNA MIR503HG可能在SSc肌成纤维细胞发育中起着重要作用[64]。lncRNA在免疫细胞中大量表达，但其作用及机制尚待进一步研究。鉴于lncRNA的重要调控作用，其在SSc中的研究将具有重要意义。

4 结 语

易感基因、表观遗传学改变和环境因素参与了疾病的发生，但易感基因的影响有限，环境因素对人类疾病发生的危险度占80%[65]。表观遗传学受环境因素的影响并可反映环境的变化，是连接环境-基因的重要枢纽。在SSc的表观遗传学研究中，以DNA甲基化和miRNA的研究最为广泛。DNA甲基化异常可通过影响SSc的血管内皮细胞、$CD4^+$T细胞和成纤维细胞的功能而参与SSc的发病机制。血清中的miRNAs与自身抗体、SSc疾病活动度和并发症密切相关，成纤维细胞中的miRNAs可直接影响成纤维细胞的活化，发挥抗纤维化或促纤维化的作用。组蛋白修饰和lncRNA在SSc中的研究尚处于起步阶段。在今后的研究中寻找SSc疾病特异的、与疾病活动度以及并发症密切相关的DNA甲基化位点或miRNAs，可为SSc的诊断、分类、并发症的预防提供重要帮助。此外，表观遗传学可应用于SSc患者对药物的治疗反应，寻找与药物效果密切相关的标志物，用药前对患者进行表观遗传学的分子标志检测可提供个体化的精准治疗。

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